Methodological Annex: A Data-Driven Global Catalogue of Occupations and Skills in Priority Sectors

1. Introduction

This annex presents the methodological foundations of a global priority sector jobs catalogue. The objective is to provide a transparent and internationally comparable approach to identifying the occupations most typically employed within priority sectors—Infrastructure and Energy, Manufacturing, Healthcare, Tourism, and Agribusiness—and to link these occupations to the skills and competencies required to sustain sectoral growth.

In the absence of harmonized global data linking industries, occupations, and skills at a detailed level, the approach relies on the United States as a benchmark economy, given the availability of rich and internally consistent labor market information. The resulting occupation profiles are subsequently validated and refined using country-level data from SAR countries, ensuring relevance beyond the benchmark context.

2. Data

The construction of the global priority sector jobs catalogue draws on three main data sources.

First, industry–occupation employment patterns are obtained from the U.S. National Employment Matrix 2024–2034, produced by the U.S. Bureau of Labor Statistics (BLS). The matrix reports employment for occupations at the 6-digit U.S. Standard Occupational Classification (SOC) level, covering 832 detailed occupations, by NAICS-based industries. Industries are reported at the highest level of detail permitted by data quality and confidentiality considerations, which implies aggregation in some cases. Self-employed workers are reported as a separate category and are not linked to specific industries; as a result, self-employment is excluded from the industry–occupation mapping, representing approximately 6 percent of total employment in the U.S. labor force.

Second, occupation-level labor market characteristics are drawn from the U.S. Occupational Outlook Handbook (2024). For each occupation, the catalogue incorporates information on typical entry-level education requirements, on-the-job training requirements, projected numbers of new jobs, and projected employment growth rates over the 2024–2034 period. In addition, detailed information on earnings is included, covering both annual and hourly wages. Wage indicators summarize the distribution of earnings within occupations using measures of central tendency and dispersion, including median wages and selected percentile-based ratios that capture within-occupation wage inequality.

Third, detailed information on skills and tasks associated with each occupation is sourced from the O*NET database. For each occupation, the catalogue incorporates standardized measures of skill importance and skill proficiency, as well as detailed task descriptions. These data provide a systematic link between occupations and the underlying skills and competencies required to perform them, allowing the jobs catalogue to move beyond occupational titles and earnings and toward a skills-based characterization of labor demand.

Finally, household survey microdata from labor force surveys (LFS) in SAR countries are used for validation and contextualization. These data are used to compute occupation-specific employment shares within priority sectors at the country level, in order to assess whether the occupations identified as typical within priority sectors in the U.S. benchmark are also prominent in other country contexts. Based on these data, country-specific occupation catalogues are constructed, including selected occupation-level indicators such as typical level of education, wages, total employment, and indicators of informality. Occupations are defined using the International Standard Classification of Occupations (ISCO) at the highest level of detail permitted by the data, ensuring consistency with sectoral definitions while preserving cross-country comparability.

3. Methodology

The methodology proceeds in three main steps: defining priority sectors at the industry level, mapping industries to occupations using employment data, and enriching occupations with skills-related information.

3.1. Priority sector definition and industry mapping

The starting point is the definition of priority sectors developed by the World Bank Global Team, which is based on economic activities classified under ISIC Rev.4. To operationalize these definitions using U.S. labor market data, priority sector definitions are translated from ISIC Rev.4 into NAICS using the 2022 NAICS–ISIC concordance.

Priority sectors are defined as follows. Infrastructure and Energy includes ISIC divisions 35–39 (electricity, gas, water supply, sewerage, waste management), 41–43 (construction), and 61 (telecommunications).[1] Manufacturing corresponds to ISIC divisions 13–33, covering all manufacturing activities excluding agribusiness-related processing. Healthcare includes ISIC divisions 86 (human health activities) and 87 (residential care activities). Tourism follows the ILO definition at the ISIC four-digit level, encompassing accommodation, food and beverage services, passenger transport, travel services, and recreation-related activities.[2] Agribusiness includes wage employment in ISIC divisions 01–03 (crop and animal production, forestry, and fishing) and all employment in ISIC divisions 10–12 (manufacture of food products, beverages, and tobacco).[3]

Because the correspondence between NAICS and ISIC classifications is not always one-to-one, some NAICS 6-digit industries map to multiple ISIC activities, which in a limited number of cases belong to more than one of the six broad sectors (the five priority sectors plus a residual category). In these cases, no more disaggregated employment information is available to allocate employment across activities. Employment is therefore distributed equally across linked ISIC industries, following a standard equal-allocation assumption used in the literature under data constraints, which applies to only 36 out of 1,012 NAICS industries.

An illustrative example is provided by NAICS 312140 (Distilleries). This NAICS industry maps to ISIC 2011 (Manufacture of basic chemicals), corresponding to the Manufacturing sector, and to ISIC 1101 (Distilling, rectifying and blending of spirits), corresponding to Agribusiness. The former captures the production of

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[1] Within Infrastructure and Energy, additional distinctions are introduced to separately identify energy activities (ISIC 35), digital infrastructure (ISIC 61), and core infrastructure (ISIC 36–39 and 41–43).

[2] Specifically, tourism includes the following ISIC Rev.4 four-digit codes: 5510, 5520, 5590, 5610, 5629, 5630, 6810, 6820, 4911, 4922, 5011, 5021, 5110, 7710, 7721, 9000, 9102, 9103, 9200, 9311, 9319, 9321, 9329, 7911, 7912, and 7990.

[3] Although self-employment is reported separately in the U.S. National Employment Matrix and is therefore excluded from the industry–occupation mapping, an additional adjustment is applied for agribusiness primary activities (ISIC divisions 01–03) to ensure consistency with the sector definition. Specifically, the agribusiness definition includes wage employment only in divisions 01–03, while all employment is included in ISIC divisions 10–12. To operationalize this definition, occupation-level employment in ISIC divisions 01–03 is scaled using sector-level wage employment shares from the 2024 Current Population Survey (CPS), Table 15. The resulting adjustment factor is 0.694, corresponding to the share of wage employment in total employment in these divisions.

non-potable ethyl alcohol, while the latter corresponds to potable alcohol production. In the absence of data allowing employment to be disaggregated between these two activities, employment in NAICS 312140 is split evenly across sectors, assigning 50 percent to Manufacturing and 50 percent to Agribusiness. Table 1 illustrates this mapping and allocation procedure.

Table 1. Example of NAICS–ISIC mapping across priority sectors: Distilleries (NAICS 312140)

(a) Concordance 2022 NAICS to ISIC Rev. 4

naics	*naics-title*	*isic*	*isic-title*	*infra_energy*	*manufacturing*	*healthcare*	*tourism*	*agribusiness*
312140	Distilleries	1101	Distilling, rectifying and blending of spirits	0	0	0	0	1
312140	Distilleries	2011	Manufacture of basic chemicals	0	1	0	0	0

(b) Priority sector definition at NAICS 6-digit level

naics	*naics-title*	*infra_energy*	*manufacturing*	*healthcare*	*tourism*	*agribusiness*
312140	Distilleries	0	0.5	0	0	0.5

Applying this approach across all NAICS industries results in a set of priority sector identifiers at the NAICS 6-digit level, expressed as sector-specific employment shares that sum to one once the residual ("rest") sector is included for each NAICS industry. These identifiers constitute the basis for linking NAICS industries to priority sectors in subsequent steps of the analysis, including the mapping of industries to occupations using the U.S. National Employment Matrix.

3.2. Occupation mapping using employment data

The second step of the methodology consists of mapping priority sectors to occupations using the U.S. National Employment Matrix 2024–2034, which reports employment by 6-digit SOC occupation across NAICS-based industries. Using the priority sector identifiers defined at the NAICS 6-digit level in the previous step, occupation-level employment is allocated to priority sectors based on industry–occupation employment relationships observed in the U.S. data.

Because industries in the National Employment Matrix are reported at the highest level of detail permitted by data quality and confidentiality constraints, employment information is not always available at the full NAICS 6-digit level. When employment is reported for aggregated NAICS industries, and these aggregated industries are linked to more than one of the six broad sectors (the five priority sectors plus a residual category), employment is distributed equally across the underlying NAICS 6-digit industries. This equal-allocation assumption mirrors the approach used in the sector mapping step and reflects the absence of more granular information. Where available, this assumption can be relaxed in future work by incorporating observed employment shares from household survey data.

Applying this procedure yields estimates of sector-specific employment shares for each 6-digit SOC occupation, which are then used to rank occupations within each priority sector according to their intensity of use. These rankings form the basis for identifying "typical" occupations within each priority sector based on U.S. employment patterns.

Although the National Employment Matrix covers 832 SOC occupations, two occupations—Patternmakers, wood and Terrazzo workers and finishers—do not report sufficient industry-level detail to allow a meaningful allocation across industries and sectors. As a result, these occupations are excluded from the sector–occupation mapping. Consequently, the U.S.-based jobs catalogue includes 830 occupations rather than the full set of 832 SOC occupations.

3.3. <u>Occupation enrichment and validation</u>

In a final step, the sector–occupation mapping is enriched with occupation-level attributes and validated using country-level data. For U.S.-based occupations, a comprehensive set of characteristics is incorporated from the U.S. Occupational Outlook Handbook (OOH) and the O*NET databases. Separate occupation catalogues are constructed for the U.S. benchmark and for each SAR country, reflecting differences in data availability and occupational classification systems.

U.S. occupation-level attributes (OOH):

For each U.S. occupation defined at the 6-digit SOC level, the catalogue includes the following fields:

- **Occupation code and occupation title**
  Six-digit SOC occupation code and corresponding occupation title.

- **Entry level of education**
  Typical level of formal education that most workers need to enter the occupation, as defined by the OOH.

- **On-the-job training**
  Occupation-specific training or preparation typically required after entry into employment for a worker to attain competency, as defined by the OOH. Categories include apprenticeships, internships or residencies, long-term, moderate-term, short-term on-the-job training, or no additional training.[4]

- **Delivery method**
  Composite indicator constructed from the interaction between entry-level education and on-the-job training requirements, summarizing the predominant mode through which skills are typically acquired. Entry-level education is grouped into no formal education, formal non-tertiary education, and formal tertiary education. On-the-job training is treated as a binary dimension, distinguishing occupations that require no on-the-job training from those that require any type of on-the-job training (including apprenticeships, internships or residencies, and long-, moderate-, or short-term on-the-job training). Combining these two dimensions yields six mutually exclusive delivery method categories: (i) none / non-formal lifelong training (no formal education and no on-the-job training); (ii) industry-led on-the-job training (no formal education and any on-the-job training); (iii) formal non-tertiary education combined with industry-led on-the-job training; (iv) formal non-tertiary education only (no on-the-job training); (v) formal tertiary education combined with industry-led on-the-job training; and (vi) formal tertiary education only (no on-the-job training).

- **Projected number of new jobs**
  Projected numeric change in employment between 2024 and 2034, as reported in the Occupational Outlook Handbook. For analytical clarity and comparability across occupations, this variable is converted into an ordered categorical indicator using the original OOH brackets: Declining (declining employment); Very low (0 to 999 new jobs); Low (1,000 to 4,999); Medium (5,000 to 9,999); High (10,000 to 49,999); and Very high (50,000 or more new jobs).

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[4] Detailed definitions of on-the-job training categories are available in the BLS Occupational Outlook Handbook glossary: https://www.bls.gov/ooh/about/glossary.htm

- **Projected growth rate**
  Projected percentage change in employment between 2024 and 2034, as defined in the Occupational Outlook Handbook. For presentation purposes, projected growth rates are reported using the standard OOH growth descriptors, which classify occupations based on the magnitude of the projected percentage change: much faster than average (increase of 7 percent or more); faster than average (increase of 5 to 6 percent); as fast as average (increase of 3 to 4 percent); slower than average (increase of 1 to 2 percent); little or no change (increase of less than 1 percent to a decrease of less than 1 percent); and decline (decrease of 1 percent or more).

- **Share of employment in the sector (U.S.)**
  Occupation-specific employment share within each priority sector, computed using the U.S. National Employment Matrix and reflecting the relative intensity of use of each occupation within sectors.

- **Median pay rank**
  Categorical indicator derived from OOH median annual wage brackets. Occupations are classified into five ordered categories based on reported median pay: Very low (less than $37,500); Low ($37,500 to $49,999); Medium ($50,000 to $74,999); High ($75,000 to $99,999); and Very high ($100,000 or more). Occupations for which annual wage information is not available are coded as N/A.

- **Annual wage indicators**
  Mean and median annual wages, together with additional distributional measures where available, derived from OEWS May 2024 data.

- **Hourly wage indicators**
  Mean and median hourly wages, together with additional distributional measures where available, computed consistently with annual wage indicators.

All wage statistics are computed for workers with positive earnings only and are included to capture both central tendencies and within-occupation dispersion in pay.

Skills and tasks (O*NET):

To complement labor market and education indicators, the catalogue incorporates detailed information on tasks and skills from the O*NET database, enabling a skills-based characterization of occupations within priority sectors.

For each occupation, the following O*NET-based fields are included:

- **Task descriptions (Task 1 to Task 65)**
  Standardized textual descriptions of the core work activities performed in each occupation, reported as up to 65 occupation-specific task statements.

- **Skill importance**
  Measures of how critical each skill is for effective job performance in the occupation, independent of the level at which it must be applied. Skill importance is reported on a standardized O*NET scale from 1 (not important) to 5 (extremely important), allowing comparisons across skills and occupations.

- **Skill proficiency (skill level)**
  Measures of the degree of competency, mastery, or complexity with which each skill must be applied in the occupation, conditional on its relevance. Skill level is reported on a standardized O*NET scale from 0 to 7, with higher values indicating greater required proficiency.[5]

The skills covered span cognitive, technical, and interpersonal domains, including Active Learning, Active Listening, Complex Problem Solving, Critical Thinking, Coordination, Judgment and Decision Making, Management of Financial, Material, and Personnel Resources, Operations Analysis, Programming, Quality Control Analysis, Reading Comprehension, Service Orientation, Social Perceptiveness, Systems Analysis, Technology Design, Troubleshooting, Writing, and related skills. Two separate skill-based catalogues are produced for U.S. occupations in priority sectors: one reporting skill importance and another reporting skill proficiency levels.

Country-level occupation catalogues and validation (SAR)

To validate U.S.-based occupation profiles and assess their relevance across country contexts, country-specific occupation catalogues are constructed using household survey microdata from labor force surveys (LFS) in SAR countries. Occupations are defined using the International Standard Classification of Occupations (ISCO) at the highest level of detail permitted by the data, implying a lower level of occupational granularity than in the U.S. SOC-based catalogue.

For each occupation in each country, the following fields are included:

- **Occupation code and occupation title (ISCO)**
  Occupation identifiers and titles defined according to the International Standard Classification of Occupations, at the most detailed level available in each survey.

- **Typical level of education**
  Typical educational attainment of workers in the occupation, defined as the median level of completed education among employed individuals in that occupation.

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[5] Skill measures are drawn from the O*NET Content Model and are based on ratings provided by trained occupational analysts, as indicated by the Domain Source = Analyst in the O*NET skills database. Analysts are provided with detailed occupational information—including task statements and standardized occupational descriptors—to facilitate the skill rating process. For each occupation–skill pair, eight trained occupational analysts independently provide ratings, and the value reported in the database corresponds to the arithmetic mean of these ratings (Fleisher and Tsacoumis, 2018).

Skill importance is rated using the question "How important is this skill to the performance of the occupation?" and recorded on a five-point ordinal scale (1 = Not important; 2 = Somewhat important; 3 = Important; 4 = Very important; 5 = Extremely important). Skill level (proficiency) is rated using the question "What level of this skill is needed to perform the occupation?" and recorded on a seven-point scale (0–7) reflecting increasing complexity, autonomy, and judgment in skill application, following the standard definitions used in the O*NET Abilities and Occupational Skills framework (Burgoyne and Reeder, 2025).

Skill level ratings are anchored to behavioral descriptions that illustrate concrete work activities at different points along the proficiency scale. These behavioral anchors are defined ex ante and applied consistently across occupations. For example, for the skill Writing, a lower-level anchor corresponds to "Write down a guest's order at a restaurant" (Level 2), while a higher-level anchor corresponds to "Write an email to staff outlining new directives" (Level 4), illustrating how proficiency levels are grounded in observable job tasks (Crawford et al., 2021).

By anchoring ratings to concrete behavioral examples and aggregating assessments across multiple trained analysts using standardized procedures, O*NET ensures internal consistency and cross-occupation comparability of both skill importance and skill level measures across technical and non-technical skill domains.

- **Share of employment in the sector**
  Occupation-specific share of total employment within each priority sector at the country level, computed using sector–occupation employment relationships observed in household survey data.

- **Median pay rank**
  Categorical indicator of relative pay, constructed using occupation-level quintiles of annual wages within each country. Occupations are classified into five ordered categories (very low, low, medium, high, and very high) based on their position in the country-specific wage distribution.

- **Annual wages**
  Mean and median annual wages for workers with positive earnings in the occupation, together with the number of observations underlying the estimates. All wage values are expressed in 2021 purchasing power parity (PPP) terms to support cross-country comparability.

- **Hourly wages**
  Mean and median hourly wages for workers with positive earnings in the occupation, together with the corresponding number of observations. Hourly wages are also expressed in 2021 PPP terms.

- **Total employment**
  Total number of employed workers in the occupation, as observed in the household survey microdata.

- **Share of self-employed workers**
  Proportion of workers in the occupation classified as self-employed.

- **Share of unpaid workers**
  Proportion of workers in the occupation engaged in unpaid employment, where identifiable in the survey data.

- **Informality rate**
  Measures of informal employment at the occupation level, constructed using country-specific definitions and variables available in the household surveys.[6]

- **Skill category**
  Broad skill classification derived from ISCO major groups, grouping occupations into high-skill (ISCO groups 1–3), medium-skill (ISCO groups 4–8), and low-skill (ISCO group 9) categories.

These country-level catalogues are used to assess whether occupations identified as typical within priority sectors in the U.S. benchmark are also prominent in other country contexts, and to examine differences in employment structure, education, and wage outcomes across countries.

[6] In the case of Bangladesh, informality is proxied by the share of wage employees without a labor contract.